

S&DS 365 / 665
Intermediate Machine Learning
Variational Inference

October 6

Yale

Reminders

- Assignment 2 due Thursday
- Midterm week from today, in class
- Material up to and including today's class
- Cheat sheet: One side of 8.5x11 sheet, handwritten
- Quiz 3 scores posted

Quiz Summary

Section Filter ▾

 Student Analysis

 Item Analysis

μ Average
Score

92%

$\nearrow$ High Score

100%

$\searrow$ Low Score

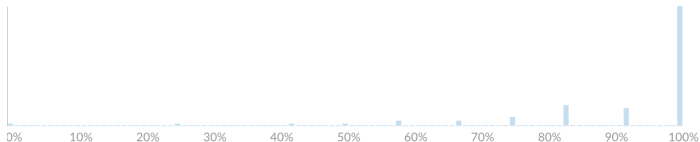
0%

σ Standard
Deviation

1.78

⌚ Average
Time

15:30



Review sessions

Keyi: Thursday, Oct 9, 4pm in KT207

Awni: Friday, Oct 10, 5pm in KT1101

Zehao: Saturday, Oct 11, 10am on Zoom

For Today

- Variational inference: The ELBO
- Derivations and examples
- Next time: Variational autoencoders (VAEs)

Welcome to the webpage of the Adaptive Bayesian Intelligence Team @ [RIKEN AIP](#).



Keynote at the [3rd Conference on Lifelong Learning Agents \(CoLLAs\) 2024](#)
[[Slides](#)]



[ICML 2024: Variational Learning is Effective for Large Deep Networks](#)



[ICLR 2023: SAM as an Optimal Relaxation of Bayes](#)

Humans, animals, and other living beings have a natural ability to autonomously learn throughout their lives and quickly adapt to their surroundings, but computers lack such abilities. Our goal is to bridge such gaps between the learning of living-beings and computers. We are machine learning researchers with an expertise in areas such as approximate inference, Bayesian statistics, continuous optimization, information geometry, etc. We work on a variety of learning problems, especially those involving supervised, continual, active, federated, online, and reinforcement learning. Please check out [research](#) and [publications](#) pages for a more exhaustive overview.

If you are interested in joining us, see the [people](#) page and the news below for current opportunities.



Recall: Inverting generative models

Template for generative model:

- 1 Choose Z
- 2 Given z , generate (sample) X

We often want to invert this:

- 1 Given x
- 2 What is Z that generated it?



Inverting models

Bayesian setup:

- 1 Choose θ
- 2 Given θ , generate (sample) X

Posterior inference:

- 1 Given x
- 2 What is θ that generated it?



Inverting models

Often we have two types of latent (hidden) variables, which we'll denote by (θ, z) .

The variable θ will typically be continuous while z is discrete.

The goal is to approximate $p(\theta, z | x)$.



Approximate inference

If we have a random vector $Z \sim p(Z \mid x)$, we might want to compute the following:

- marginal probabilities $\mathbb{P}(Z_i = z \mid x)$
- marginal means $\mathbb{E}(Z_i = z \mid x)$
- most probable assignments $z^* = \arg \max_z \mathbb{P}(\{Z_i = z_i\} \mid x)$
- maximum marginals $z_i^* = \arg \max_{z_i} \mathbb{P}(Z_i = z_i \mid x)$
- joint probability $\mathbb{P}(Z \mid x)$
- joint mean $\mathbb{E}(Z \mid x)$

Each of these quantities is intractable to calculate exactly, in general.

Gibbs sampler

Generic algorithm:

- 1 Sample one component holding others fixed
- 2 Iterate

Gibbs sampler

Generic algorithm:

- 1 Choose index j randomly
- 2 Sample θ_j fixing $\theta_i, i \neq j$ and x
- 3 Iterate

Gibbs sampler

Generic algorithm:

- 1 Choose index j randomly
- 2 Sample θ_j from $\mathbb{P}(\theta_j \mid \theta_i, i \neq j, x)$.
- 3 Iterate

When run for sufficiently long, this will draw a sample from the posterior $\mathbb{P}(\theta \mid x)$.

Variational methods

- Gibbs sampling is *stochastic* approximation
- Variational methods iteratively make *deterministic* approximations
- Variational and Markov chain approximations originated in physics

Stochastic approximation–Ising model

Gibbs sampler

Iterate until converged:

- 1 Choose vertex $s \in V$ at random
- 2 Sample z_s holding others fixed

$$\theta_s = \text{sigmoid} \left(\beta_s + \sum_{t \in N(s)} \beta_{st} z_t \right)$$
$$Z_s \mid \theta_s \sim \text{Bernoulli}(\theta_s)$$

You should verify that this is the correct conditional distribution of Z_s holding the others fixed.

Deterministic approximation

Mean field variational algorithm

Iterate until converged:

- 1 Choose vertex $s \in V$ at random
- 2 Update mean μ_s holding others fixed

$$\mu_s = \text{sigmoid} \left(\beta_s + \sum_{t \in N(s)} \beta_{st} \mu_t \right)$$

Deterministic vs. stochastic approximation

- The z_s variables are random
- The μ_s variables are deterministic
- The Gibbs sampler convergence is in distribution
- The mean field convergence is numerical
- The Gibbs sampler approximates the full distribution
- The mean field algorithm approximates the mean of each node

Example 2: A finite mixture model

Fix two distributions F_0 and F_1 , with densities $f_0(x)$ and $f_1(x)$, and form the mixture model

$$\begin{aligned}\theta &\sim \text{Beta}(\alpha, \beta) \\ X | \theta &\sim \theta F_1 + (1 - \theta) F_0.\end{aligned}$$

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The likelihood for data $x_1, \dots, x_n$ is

$$p(x_{1:n}) = \int_0^1 \text{Beta}(\theta | \alpha, \beta) \prod_{i=1}^n (\theta f_1(x_i) + (1 - \theta) f_0(x_i)) d\theta.$$

Our goal is to approximate the posterior $p(\theta | x_{1:n})$

Stochastic approximation

Gibbs sampler

- 1 Sample $Z_i \mid \theta, x_{1:n}$ for $i = 1, \dots, n$
- 2 Sample $\theta \mid z_{1:n}, x_{1:n}$

The first step is carried out by sampling

$$Z_i = \begin{cases} 1 & \text{with probability } \propto \theta f_1(x_i) \\ 0 & \text{with probability } \propto (1 - \theta) f_0(x_i) \end{cases}$$

Stochastic approximation

Gibbs sampler

- 1 Sample $Z_i \mid \theta, x_{1:n}$ for $i = 1, \dots, n$
- 2 Sample $\theta \mid z_{1:n}, x_{1:n}$

The second step is carried out by sampling

$$\theta \sim \text{Beta} \left(\sum_{i=1}^n z_i + \alpha, n - \sum_{i=1}^n z_i + \beta \right).$$

Posterior over θ is approximated as *mixture* of Beta distributions;
number of components is $n + 1$

Variational inference: Strategy

- We'd like to compute $p(\theta, z | x)$, but it's too complicated.
- Strategy: Approximate as $q(\theta, z)$ that has a “nice” form
- q is a function of “variational parameters” optimized for *each* x .
- Maximize a lower bound on $p(x)$.

Variational inference: The ELBO

The ELBO is the following lower bound on $\log p(x)$:

$$\log p(x) = \int \sum_z q(z, \theta) \log p(x) d\theta$$

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We maximize this over the parameters of q



Variational inference: The ELBO

The inequality above uses concavity of the logarithm:

$$\log \left(\sum_{\alpha} w_{\alpha} x_{\alpha} \right) \geq \sum_{\alpha} w_{\alpha} \log x_{\alpha}$$

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So, if $q_{\alpha} \geq 0$ and $p_{\alpha} \geq 0$ sum (or integrate) to one, then

$$0 = \log \left(\sum_{\alpha} p_{\alpha} \right) = \log \left(\sum_{\alpha} q_{\alpha} \frac{p_{\alpha}}{q_{\alpha}} \right) \geq \sum_{\alpha} q_{\alpha} \log \left(\frac{p_{\alpha}}{q_{\alpha}} \right)$$

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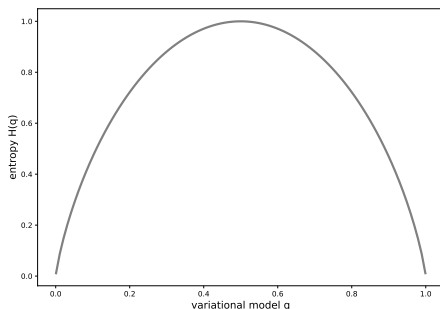
Therefore

$$\sum_{\alpha} q_{\alpha} \log \left(\frac{q_{\alpha}}{p_{\alpha}} \right) \geq 0$$

Variational inference: The ELBO

The ELBO is $H(q) + \mathbb{E}_q(\log p)$

The entropy term $H(q)$ encourages q to be spread out:



The cross-entropy $\mathbb{E}_q \log p$ tries to match q to p

Variational inference: Strategy

- Choose a simple for the variational approximate posterior $q_{\gamma}(\theta, z | x)$.
- Optimize the ELBO over the parameters γ

Example 2: A finite mixture model

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Our goal is to approximate the posterior $p(\theta | x_{1:n})$

Variational approximation

Our variational approximation is

$$q(z, \theta) = q(\theta \mid \gamma_1, \gamma_2) \prod_{i=1}^n q_i^{z_i} (1 - q_i)^{(1-z_i)}$$

where $q(\theta \mid \gamma_1, \gamma_2)$ is a $\text{Beta}(\gamma_1, \gamma_2)$ distribution, and $0 \leq q_i \leq 1$ are n free parameters.

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Need to maximize ELBO $H(q) + \mathbb{E}_q \log p$

Let's sketch part of the calculation

Variational approximation

First, we have

$$\log p(x, \theta, z) = \log p(\theta \mid \alpha, \beta) + \sum_{i=1}^n \left\{ \log (\theta f_1(x_i))^{z_i} + \log ((1 - \theta) f_0(x_i))^{(1-z_i)} \right\}$$

Variational approximation

Next, take expectation with respect to q and use identities such as

$$\mathbb{E}_q \log \theta = \psi(\gamma_1) - \psi(\gamma_1 + \gamma_2)$$

for the digamma function $\psi(\cdot)$.

Variational approximation

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for the digamma function $\psi(\cdot)$.

So, for example,

$$\begin{aligned}\mathbb{E}_q \log (\theta f_1(x_i))^{z_i} &= q_i \mathbb{E}_q \log \theta + q_i \log f_1(x_i) \\ &= q_i (\psi(\gamma_1) - \psi(\gamma_1 + \gamma_2)) + q_i \log f_1(x_i)\end{aligned}$$

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After some calculus and algebra, we end up with the following algorithm (see the notes on the course web page for more detail)

Variational algorithm for mixture

Variational inference

Iterate the following steps for variational parameters $q_{1:n}$ and (γ_1, γ_2) :

- 1 Holding q_i fixed, set $\gamma = (\gamma_1, \gamma_2)$ to

$$\gamma_1 = \alpha + \sum_{i=1}^n q_i \quad \gamma_2 = \beta + n - \sum_{i=1}^n q_i$$

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$$\gamma_1 = \alpha + \sum_{i=1}^n q_i \quad \gamma_2 = \beta + n - \sum_{i=1}^n q_i$$

- 2 Holding γ_1 and γ_2 fixed, set q_i to

$$q_i = \frac{f_1(x_i) \exp \psi(\gamma_1)}{f_1(x_i) \exp \psi(\gamma_1) + f_0(x_i) \exp \psi(\gamma_2)}$$

Variational algorithm for mixture

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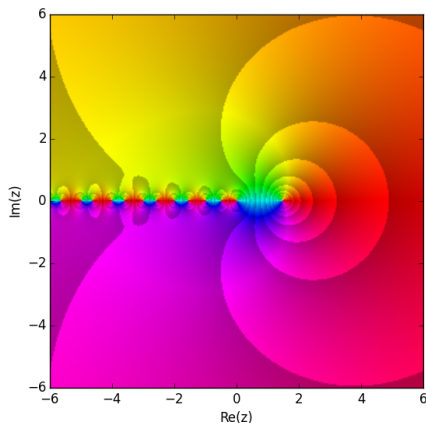
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$$q_i = \frac{f_1(x_i) \exp \psi(\gamma_1)}{f_1(x_i) \exp \psi(\gamma_1) + f_0(x_i) \exp \psi(\gamma_2)}$$

After convergence, approximate posterior distribution over θ is

$$\hat{p}(\theta \mid x_{1:n}) = \text{Beta}(\theta \mid \gamma_1, \gamma_2)$$

Digamma function



$\psi(x)$ is the *digamma function*

https://en.wikipedia.org/wiki/Digamma_function

Deterministic approximation

- Convergence is numerical, not stochastic
- Posterior is approximated as a *single* Beta
- Very similar algorithm is used for topic models

Example 3: More general mixtures

$$\theta \sim \text{Dirichlet}(\alpha_1, \dots, \alpha_k)$$

$$X | \theta \sim \theta_1 F_1 + \dots + \theta_k F_k$$

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$$\theta \sim \text{Dirichlet}(\alpha_1, \dots, \alpha_k)$$

$$X | \theta \sim \theta_1 F_1 + \dots + \theta_k F_k$$

The likelihood for single data point x is

$$p(x) = \int \text{Dirichlet}(\theta | \alpha_1, \dots, \alpha_k) \left(\sum_{j=1}^k \theta_j f_j(x) \right) d\theta.$$

When distributions F_j are learned, this is a “topic model.” Variational inference is one of the most useful ways of training topic models

- Notes on variational methods: Please read Sections 1–4
- The other sections are more advanced / specialized material
- Next up: Variational autoencoders
- VAEs not on midterm, but please review basics of variational methods discussed so far, including the ELBO

Some questions about variational inference

Variational inference

Q: What is the best q we could use?

Variational inference

Q: What is the best q we could use?

A: The true posterior $q(z, \theta | x) = p(z, \theta | x)$

Variational inference

Q: What is the best q we could use?

A: The true posterior $q(z, \theta | x) = p(z, \theta | x)$

Why? Because this maximizes the ELBO. Mathematically,

$$\sum_z \int q(z, \theta) \log \left(\frac{q(z, \theta)}{p(z, \theta | x)} \right) d\theta = 0$$

in this case, so the ELBO inequality is an equality.

Variational inference

Q: How does the ELBO regularize?

Variational inference

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A: The entropy term favors distributions that are “spread out”

Variational inference

Q: How does the ELBO regularize?

A: The entropy term favors distributions that are “spread out”

Why? For discrete distributions the maximum entropy distribution is uniform. For Gaussian, the entropy is $\log \sigma^2$ which favors σ^2 large.

Variational inference

Q: Is the ELBO easy to maximize?

Variational inference

Q: Is the ELBO easy to maximize?

A: No.

Variational inference

Q: Is the ELBO easy to maximize?

A: No.

Why? In general it is non-convex, and the solution depends on where we start an iterative algorithm. This is unlike the Gibbs sampler, which converges to the right thing if we wait long enough.

Next class: Variational autoencoders

- Variational autoencoders are generative models that are trained using variational inference
- The “decoder” is a neural net that generates from a latent variable
- The “encoder” approximates the posterior distribution with another neural network trained using variational inference

Summary

- Gibbs sampling makes stochastic approximations
- Variational methods make deterministic approximations
- General recipe: Maximize ELBO over variational parameters
- Gives a powerful approach to generative modeling